RESPONSIVE PORTFOLIO WEBSITE

CS Web Development Midterm Project Documentation Report

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GitHub Repository	`github.com/jjaguilarf05-sketch/responsive-portfolio-midterm`	**Live Site URL**	`jjaguilarf05-sketch.github.io/responsive-portfolio-midterm`

1. PORTFOLIO ENHANCEMENTS (BEFORE & AFTER)

The objective of this midterm project was to transform an initial single-page HTML design into a fully functional, structured, and responsive personal portfolio. Key updates focused on code modularity, design polish, and robust internal navigation.

- **Code Separation:** Split inline CSS styles out of `index.html` into an external `style.css` file connected via standard `<link rel="stylesheet">` tag.
- **Asset Pathing Fixes:** Resolved broken logo references by standardizing image assets under the local `assests/` directory (e.g., `assests/https://lh3.googleusercontent.com/gg-dl/AHiVA1nKPR7cpT0VzH5UO1lcbG_74Jk_xT2VzEGVTpK3l1sL73gBwzqpg9r1qhthpgWJXZHPZURaGXBI6msCgYqGxz4tfNUtSxz5BfeX7Zk5yZL70a5X9J56i_fIguMH5wnFEVcdh1fNyf91MhPyoroCDKRgefPMoNaa54h_1VYo9DpMf0P5_w`).
- **Complete Anchor Navigation:** Implemented missing smooth-scrolling target sections (`#about` and `#contact`) matching all top navbar items.
- **Functional Social & Project Links:** Replaced all `href="#"` placeholders with authentic repository URLs and direct project links.

FIGURE 1.1: BRAND LOGO ASSET (HTTP
DL/AHIVA1NKPR7CPT0VZH5UO1LCBG_74JK_XT2VZEGVTPK3L1SL73GBWZQPG9R1QHTHPGWJXZHPZURAGXBI6MSCGYQGX



2. FLEXBOX IMPLEMENTATION DETAILS

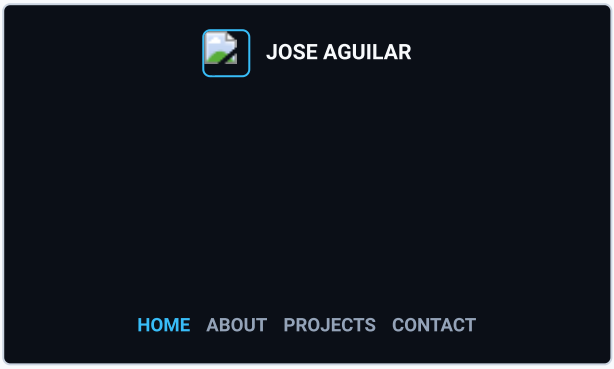
CSS Flexbox (Flexible Box Layout) was utilized across primary structural containers requiring 1D alignment, such as the top navigation bar, section items, and footer controls.

- `display: flex` — Establishes a flex container context for immediate child elements.
- `justify-content: space-between` — Spreads out branding items to the far left and navigation links to the far right.
- `align-items: center` — Ensures vertical centering alignment between logo images and text labels.
- `gap: 24px` — Provides consistent spacing between flex items without requiring margin tweaks.
- `flex-direction: column` — Used in media queries to stack navigation items vertically on small mobile displays.

FIGURE 2.1: NAVBAR CSS SNIPPET

```
.navbar { display: flex; justify-content: space-between; align-items: center; padding: 24px 0; } .nav-links { display: flex; gap: 24px; }
```

FIGURE 2.2: FLEX NAVIGATION HEADER RENDER



3. CSS GRID IMPLEMENTATION DETAILS

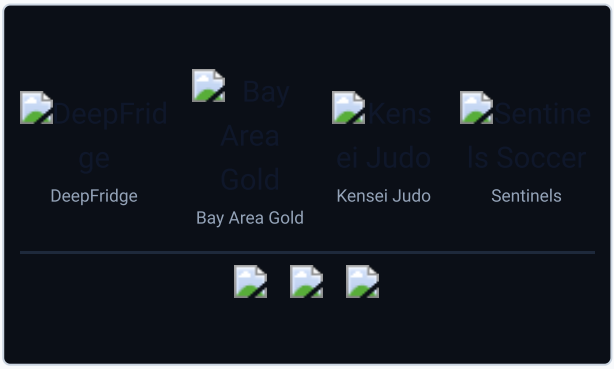
CSS Grid was implemented for the **Projects Section** gallery layout to handle complex 2D column and row arrangements dynamically without relying on fixed pixel widths.

- `display: grid` — Activates the 2D grid formatting context for project card wrappers.
- `grid-template-columns: repeat(auto-fit, minmax(200px, 1fr))` — Creates a fluid multi-column grid where cards automatically fill row space, resizing dynamically down to a 200px minimum threshold before wrapping.
- `gap: 20px` — Controls exact gutter spacing horizontally and vertically between project cards.

FIGURE 3.1: GRID CSS SNIPPET

```
.project-grid { display: grid; grid-template-columns: repeat(auto-fit, minmax(200px, 1fr)); gap: 20px; }
```

FIGURE 3.2: GRID PROJECT GALLERY ICONS & FOOTER SOCIALS



4. RESPONSIVE WEB DESIGN & MEDIA QUERIES

Media queries ensure the user interface scales comfortably across screen dimensions:

- **Desktop Viewport (> 800px):** Full horizontal Flexbox navbar, 4-column Grid layout for project cards.
- **Tablet Breakpoint (800px):** Stacks the navbar into a centered column; adjusts the project grid to a comfortable 2-column format.
- **Small Device Breakpoint (550px):** Adjusts work experience rows to stack vertically so long role titles do not overflow.
- **Mobile Breakpoint (400px):** Switches project cards to a single 1-column layout and stacks nav links vertically for tap target accessibility.

5. WRITTEN REFLECTION

> Building this portfolio helped me understand when to use Flexbox versus CSS Grid. I chose Flexbox for 1D layouts like the navbar and footer where I needed elements spaced out horizontally. I used CSS Grid for the projects section because auto-fit and minmax() allowed cards to re-arrange into columns naturally.
>
> One challenge I ran into was fixing broken image links caused by folder naming mismatches (`assests/` vs `assets/`). Additionally, separating CSS into an external file made the codebase much easier to manage. Testing across tablet and mobile breakpoints ensured text remained readable on small screens.

6. REPOSITORY & DEPLOYMENT GUIDE

Follow these steps to deploy and test the project repository:

1. Clone the repository: `git clone https://github.com/jjaguilarf05-sketch/responsive-portfolio-midterm.git`
2. Verify directory contents contain `index.html`, `style.css`, and the `assests/` image folder.
3. Deploy to GitHub Pages: Navigate to *Settings* > *Pages*, select the `main` branch root, and save.